Part 1: Foundations and Data for the Wyckoff Model

1. Project Overview (The Wyckoff Objective) While predicting macroscopic properties (like formation energy) from composition is highly useful, actually *generating* new materials requires structural knowledge. State-of-the-art materials discovery frameworks (such as Wren, WyckoffDiff, and ShotgunCSP-GW) emphasize that using **Wyckoff positions** is an elegant, coordinate-free way to represent crystal symmetry. The objective of this specific model is to predict the set of occupied Wyckoff letters for a given chemical composition and space group. This bridges the gap between a simple chemical formula and a full 3D crystal structure, acting as a crucial prior for generative materials design.

2. Dataset Used

- **Source & Size:** The model was trained on a comprehensive dataset of crystal structures (`wyckoff_structure_dataset.csv`) initially containing 210,579 instances.
- **Preprocessing & Filtering:** To ensure statistical robustness, rare space groups were filtered out. Only space groups containing more than 100 samples were retained, reducing the unique space groups from 230 down to 128, leaving a cleaned dataset of 206,606 rows. Invalid chemical formulas were also dropped.

3. Feature Engineering & Target Encoding

- **Features (260 total):** The input features were constructed by concatenating 132 physicochemical **Magpie features** (e.g., `avg_dev Electronegativity`, `mean Column`, `mean NdValence`) with 128 **One-Hot Encoded Space Group** variables (e.g., `SG_225`, `SG_47`).
- **Target Variables (Multi-Label):** Because a single material can have atoms occupying multiple different Wyckoff positions simultaneously, this was framed as a **Multi-Label Classification** problem. The `wyckoff_letters` column was parsed and binarized using `MultiLabelBinarizer` into 27 distinct classes (ranging from 'A' and 'a' through 'z').

Part 2: Methodology and Setup

4. Model Architecture (Structure-Aware Ensemble) To handle the multi-label nature of the data, the project employed a robust ensemble of gradient-boosted decision trees:

- **LightGBM (LGBM) Classifiers:** An independent classifier was trained for each of the 27 Wyckoff classes using 300 estimators, a `max_depth` of 7, a learning rate of 0.05, and a `class_weight="balanced"` strategy to account for rare Wyckoff letters.

- **XGBoost (XGB) Classifiers:** A parallel set of 27 classifiers was trained using 400 estimators, a `max_depth` of 6, and a logistic binary objective (`binary:logistic`).
- **Ensemble Fusion:** The final model predictions were generated by taking the arithmetic mean of the output probabilities from the LGBM and XGBoost models.

5. Training Procedures

- The dataset was split using an 80/20 train-test split (seed 42), resulting in 41,320 validation samples.
- The multi-label evaluation threshold was set at a standard 0.5 probability.

Part 3: Experimental Results and Visualizations

6. Numerical Performance Metrics The ensemble framework demonstrated strong predictive power for identifying symmetry positions:

- **Micro-F1 Scores:** The XGBoost model achieved a Micro-F1 of **0.851**, while the LightGBM model achieved **0.820**.
- **Retrieval Performance (Recall@K):** Because predicting the *exact* combination of letters is difficult, top-K retrieval is an essential metric. The ensemble model successfully placed the true Wyckoff letters in its top predictions with high frequency:
  - **Recall@1:** 0.49
  - **Recall@3:** 0.76
  - **Recall@5:** 0.88
  - **Recall@8:** 0.96 (Almost all true letters are found within the top 8 guesses).

7. Visual Insights & Model Diagnostics Your uploaded visualizations provide deep insights into how the model behaves across different symmetries:

- **Wyckoff Multi-label Prediction (Class Distribution & Recall@K):** The bar chart reveals severe class imbalance (the letter 'a' has over 10,000 instances, while 'y' and 'z' have fewer than 10). The overlaid Recall@K curve proves that despite this imbalance, the model rapidly recovers the correct letters within a handful of predictions.
- **Per-class Metrics (Precision, Sensitivity, F1 Heatmaps):** The heatmaps visually demonstrate that majority classes (like 'a', 'b', 'c', 'd') have near-perfect Precision and F1 scores (dark blue/dark red), while minority classes struggle with recall due to data sparsity.
- **Prediction Calibration (Reliability Diagram):** The Expected Calibration Error (ECE) plots show that the model is highly calibrated. For example, the most common class ('a') has an excellent ECE of ~0.05. The curves tightly hug the "Perfect calibration" diagonal, meaning when the model says it is 80% confident, it is correct about 80% of the time.
- **Prediction Confidence & Entropy:** A critical plot shows "Entropy: Correct vs Incorrect Predictions." Correct predictions are overwhelmingly clustered near 0.0 nats (absolute

certainty), whereas incorrect predictions are widely spread across higher entropies (1.5 to 2.5 nats). This proves the model possesses a high degree of self-awareness regarding its own errors.
- **Feature Importance:** The top 30 Magpie feature importances reveal that `avg_dev Electronegativity`, `SG 47`, `mean Column`, and `mean GSbandgap` are the dominant driving factors in determining which Wyckoff positions atoms will occupy.
- **t-SNE of Validation Set:** The 2D projection of the validation set (colored by the dominant predicted Wyckoff letter) shows distinct spatial clustering. This confirms the model has learned meaningful, separable chemical-symmetry representations without explicit 3D structural data.

Part 4: Contextualization and Conclusion

8. Comparison With Existing Literature

- **ShotgunCSP-GW:** Your methodology heavily mirrors the *ShotgunCSP-GW* paper, which uses Random Forest regressors alongside XenonPy compositional descriptors to predict Wyckoff letters. Similar to their findings (where KL divergence showed high predictability of Wyckoff letters from composition), your ensemble confirms that stoichiometry combined with space group dictates structural symmetries with high accuracy (Micro-F1 ~0.85).
- **Generative Frameworks (Wren, WyckoffDiff, WyFormer, CGWGAN):** These benchmark papers aim to bypass the massive computational cost of Density Functional Theory (DFT) by using Wyckoff positions to construct "coarse-grained" materials representations. Your model acts as the perfect precursor to these frameworks. By successfully predicting the exact Wyckoff letters (Recall@5 = 0.88), your model dramatically shrinks the search space, allowing downstream models (like WyckoffDiff or CGWGAN) to generate 3D coordinates solely for highly probable symmetry configurations.

9. Limitations and Future Work

- **Limitations:** The `Wyckoff_Letter_Fidelity` plot (Frequency Ratio) reveals that the model severely under-predicts rare Wyckoff letters (like 'l' or 'q') and slightly over-predicts common ones (like 'a' and 'j'). The model is fundamentally constrained by the extreme class imbalance inherent in crystallographic databases.
- **Future Work:** As suggested by the *WyFormer* and *Matra-Genoa* papers, future iterations could integrate a Transformer-based self-attention mechanism specifically tuned for discrete sequences, allowing the model to not just predict independent Wyckoff letters, but to predict the *auto-regressive sequence* of Wyckoff positions (i.e., learning that if position 'a' is filled by Ti, position 'b' is highly likely to be filled by O).